

Python Data Visualization Assignment Report

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Contents

1	Introduction	2
1.1	Methodology	2
1.2	Sources	2
1.3	AI Usage	2
2	Dataset	2
2.1	Analysis	3
2.2	Processing	3
2.3	Research Questions	4
2.3.1	Question 1: Activity Composition	4
2.3.2	Question 2: Geographic Patterns	4
2.3.3	Question 3: Weekly Activity Assessment	4
3	Visualization Prep	4
3.1	Storytelling	4
3.2	Colors and Themes	4
3.3	Library Choices	5
4	Visualizations	5
4.1	Q1 - Recorded Activity Types	5
4.2	Q2 - Maps	6
4.2.1	Q2a: Activities by Autonomous Community	6
4.2.2	Q2b: Activities by Municipality (Madrid)	7
4.3	Q3 - Trends	7
4.3.1	Q3a: Overview Heatmap	8
4.3.2	Q3b: Year-Specific Trends	9
4.3.3	Q3c: 2026 Weekly Activity Trend	10
4.3.4	Visualization Summary	11
5	Conclusions	11
6	References	11

1 Introduction

The aim of this project is to explore data visualization using Python, from selecting the data all the way to telling a story through it's visualization.

1.1 Methodology

The project itself followed the following process to achieve the previously mentioned goal:

1. Dataset selection
2. Data analysis and processing
3. Questions derived from the dataset
4. Visualization types needed to answer each question and storytell
5. Further data processing needed for each visualization
6. Library selection for each visualization
7. Package visualizations within a combined story narrative that tells a story about the data

1.2 Sources

Numerous sources are used throughout the project such as:

- AI
- Documentation
- Forums
- Literature
- Lecture notebooks

Regardless of the source, these are indicated through bibliography/references or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage

A point was made to use AI in this project in a manner in which it's answers are scrutinized and reviewed, to make sure hallucinations are not taken into account and answers are not just implemented without a second thought. This way, an active usage of AI is incentivized with certain restraints in place.

AI being used for this project was also hosted locally, using smaller models to do so to encourage reproducibility (since prompts are also included within the notebook and report).

2 Dataset

Data for this project was required to not have been selected by a different student. To minimize the risk of this happening, activity from a Garmin Forerunner 245 was chosen, reducing the chances of another student having the same dataset. The data located in `./data/activities.csv` consists of:

- **Shape:** 31 columns & 180 rows
- **Date span:** 21 February 2021 - 30 March 2026 (5 years)
- **Data types:** Dates, numerics, categorical, geospatial (Title contains locations)

- **Key columns:** Date, Activity Type, Time, Distance, Avg HR
- **Activity types:** Running, Walking, Cycling, Strength, Swimming, etc.

Having 31 columns allows for many different questions to be asked about the dataset, specially since some of these columns allow for other data fields to be extracted (activity intensity from Avg HR). The limited number of rows (180) whilst still being a decent amount, allows it's content to be analyzed in a more hands on way, being able to be more critical about each specific row and actively engage in processing in a more granular fashion. The dates spanning 5 years back allows for a story to be told through time, and analyze factors such as consistency. Lastly, varying data types allows for distinct types of visualizations that require specific units to be used such as coordinates (from location).

2.1 Analysis

Before asking questions about the dataset, it was further analyzed using both the `df.info()` function to check how the data was imported into the pandas dataframe. A small AI analysis was invoked using qwen3.5:9b¹ for the prompt which was the following:

Read the linked CSV file and analyze it, create a table with the following fields:

- N: Column number
- Column: Column name
- Description: Content description (what the data in the column usually) looks like), excluding null indicators
- Null indicator: Indicator that seems to be used for when the data is null
- Data type: Suggested data type to be used when reading CSV in Pandas.

The table generated was then used as a starting point although it's content was scrutinized and interpreted correctly, assessing whether or not the AI had hallucinated when generating it. Some important conclusions reached with the help of this analysis (done in section 2.1 of `data_viz_notebook.ipynb`) where:

- Some columns weren't really needed or used, such as *Favorite*
- "-" was used as a null indicator on most columns
- Correct data types for each column
- Some columns belonged to specific activity types, such as *Avg. Swolf* for swimming

2.2 Processing

Key processing steps (section 3 of `data_viz_notebook.ipynb`):

- Replaced '-' nulls with NaN
- Converted Time/Distance to numeric (regex parsing)
- Extracted locations from Title
- Created WHO-equivalent activity minutes² from Avg HR thresholds
- Weekly aggregation for time series analysis
- Activity Type simplification (treadmill running, trail running, running are all running)

Libraries: Pandas,³ NumPy,⁴ re⁵ (regex)

Some of the data processing was effectuated in order to satisfy the data requirements of some of the visualizations, which will be explained in the following section.

2.3 Research Questions

Three questions were formulated based on data exploration:

1. What types of activities dominate the dataset and their frequencies?
2. How do activity patterns vary by location?
3. Do recorded activities showcase "enough" weekly exercise?

2.3.1 Question 1: Activity Composition

Question 1 is a good starting point because it allows for the superficial exploratory analysis of the data. This facilitates the understanding of what activity types end up shaping most of the data, and is a good starting point that is taken into account when further exploring other questions.

2.3.2 Question 2: Geographic Patterns

Question 2 proceeds naturally from question 1, as it further dives into understanding the nature of the data, in this case, classifying it into geographic location rather than categorical activity types. Furthermore, it also enables the use of different visualization methods due to the unique geographic data type, location.

2.3.3 Question 3: Weekly Activity Assessment

Question 3 is purposefully subjective to some degree, to allow for a variety of approaches to visualizing and analysing the data, alongside more liberty when it comes to choosing what narrative to pursue in storytelling. This allows for many different charts to be used, and for data to be used in creative ways.

3 Visualization Prep

Before coding out the visualizations, some prep was done in order to steer those visualizations in a specific direction, and have a clear idea of what to aim for.

3.1 Storytelling

A Garmin user tracks 5+ years of fitness data. Q1^{2.3.1} and Q2^{2.3.2} reveal activity variety (locations and types). Q3^{2.3.3} dives into health trends. We follow them through their health journey.

3.2 Colors and Themes

Colors and themes were chosen purposefully in order to make charts as simple and readable as possible. The following design choices were made on purpose to enhance clarity, readability and intent:

- Green theme (emerald-500 tailwind⁶ palette) + teal accents, fitness/health connotation, high contrast.
- Minimal ink: No grid/borders, no clutter, focus on data.
- Distinct category colors for pre-attentive processing (green vs. gray).
- Custom colormaps (sequential for magnitude, diverging for trends).

These decisions were made in line with theory from the first unit.

3.3 Library Choices

Due to multiple libraries being used for this project, the following libraries were chosen pre emptively:

- **Q1**^{2.3.1}: Matplotlib⁷ (bars)
- **Q2**^{2.3.2}: Plotly⁸ (maps)
- **Q3**^{2.3.3}: Seaborn,⁹ ipywidgets,¹⁰ matplotlib⁷ (trends)

Thus laying out the following: Q1 for a basic chart, Q2 for diving into map charting, Q3 for exploring more creative types of charts.

4 Visualizations

Shared across visualizations, a color palette is declared with colors mentioned in section 3.2, alongside an `apply_theme()` function that helps consolidate themeing across multiple functions.

4.1 Q1 - Recorded Activity Types

The following options were considered: vertical bar chart, horizontal bar chart, pie chart. The horizontal bar chart option was chosen due to being able to be ordered and have the highest count at the top, making it very easy to read. The other option would have been a pie chart with the walking and running segments exploded, but that wouldn't have been appropriate considering that they took up such a huge percentage of the total activity mix.

For the dataset for this chart, data had to be aggregated to get counts by activity type, creating another data set in doing so.

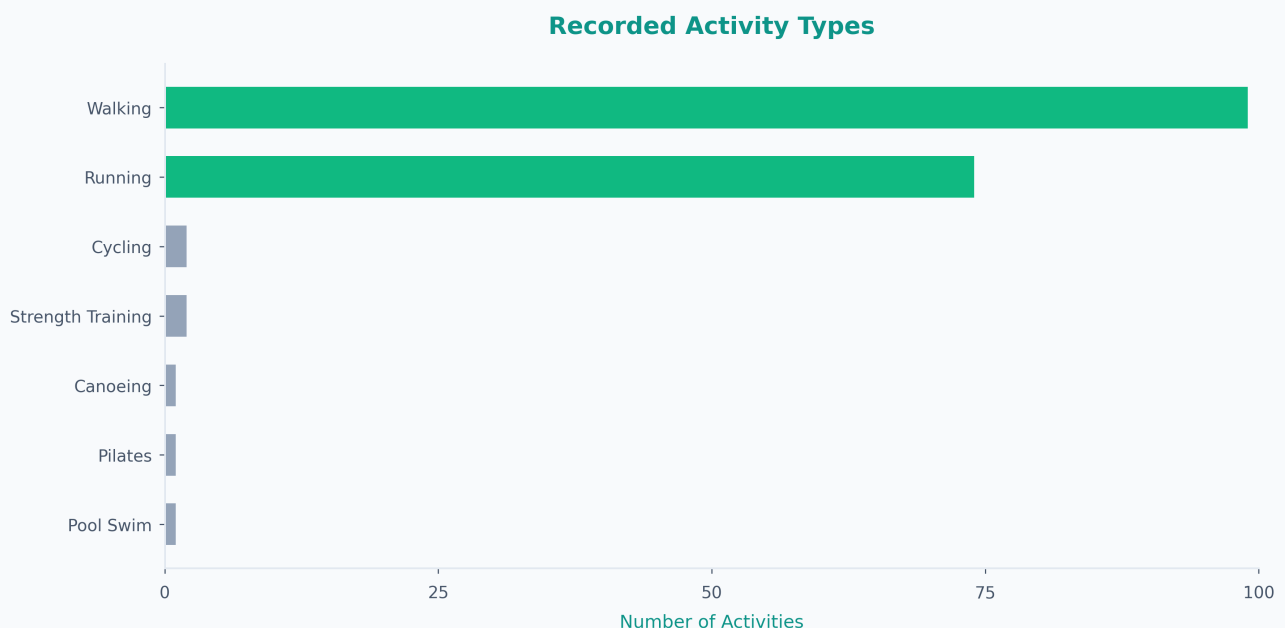


Figure 1: Q1 Chart - Recorded Activity Types

Insights: In Fig. 1 one can tell that two specific activities dominate the activity mix landscape (being walking and running). This is made even clearer with the green color making them pop out whilst the grey color make the other activities fall to the background. Styling is simple, with the chart not having many distracting elements.

4.2 Q2 - Maps

Maps were chosen for the second question to display variability in locations for recorded activities. Two maps were chosen, due to most activities being recorded in Madrid, but some activities being recorded elsewhere in Spain. This allows for two different types of map to be used.

For both maps, activities were aggregated by location, such as autonomous community or municipality, creating a special data set per chart to be used. Location itself was extracted from title, so even though it wasn't explicitly given, a work around was used to be able to obtain it.

4.2.1 Q2a: Activities by Autonomous Community

Very simple **choropleth map** that showcases the stark difference in activities recorded in Madrid vs. other autonomous communities. The choropleth map was chosen due to communities being able to be colored to show case different amounts of activities recorded, although a scatter map could have worked too.

Activities by Autonomous Community

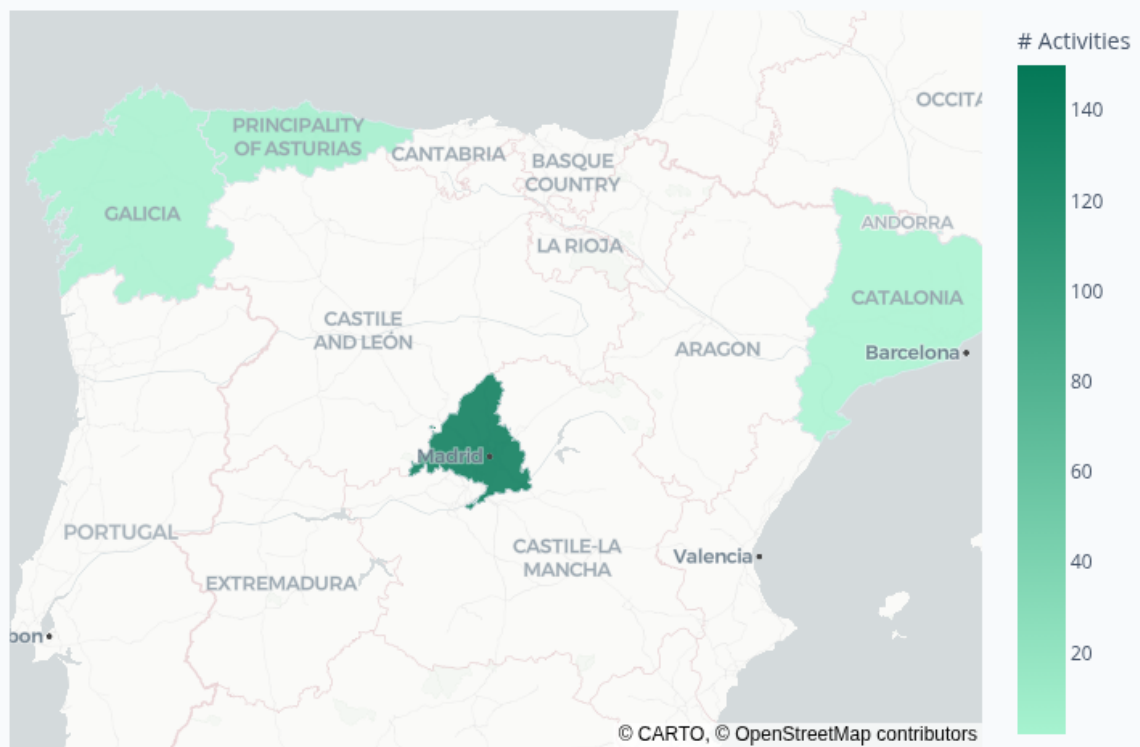


Figure 2: Q2a Chart - Activities by Autonomous Community

Insights: Fig. 2 shows a much darker shade of green for Madrid than for other autonomous communities due to Madrid being more than 90% of activities recorded.

4.2.2 Q2b: Activities by Municipality (Madrid)

Due to Fig 2 showcasing a stark difference between activities recorded in Madrid vs. other autonomous communities, it follows that an analysis should be made of activities recorded in Madrid.

A **scatter map** fits well for this analysis due to circles showing how many activities are recorded via size and color, making bigger circles stand out for the viewer easier.

Activities by Municipality (Madrid)

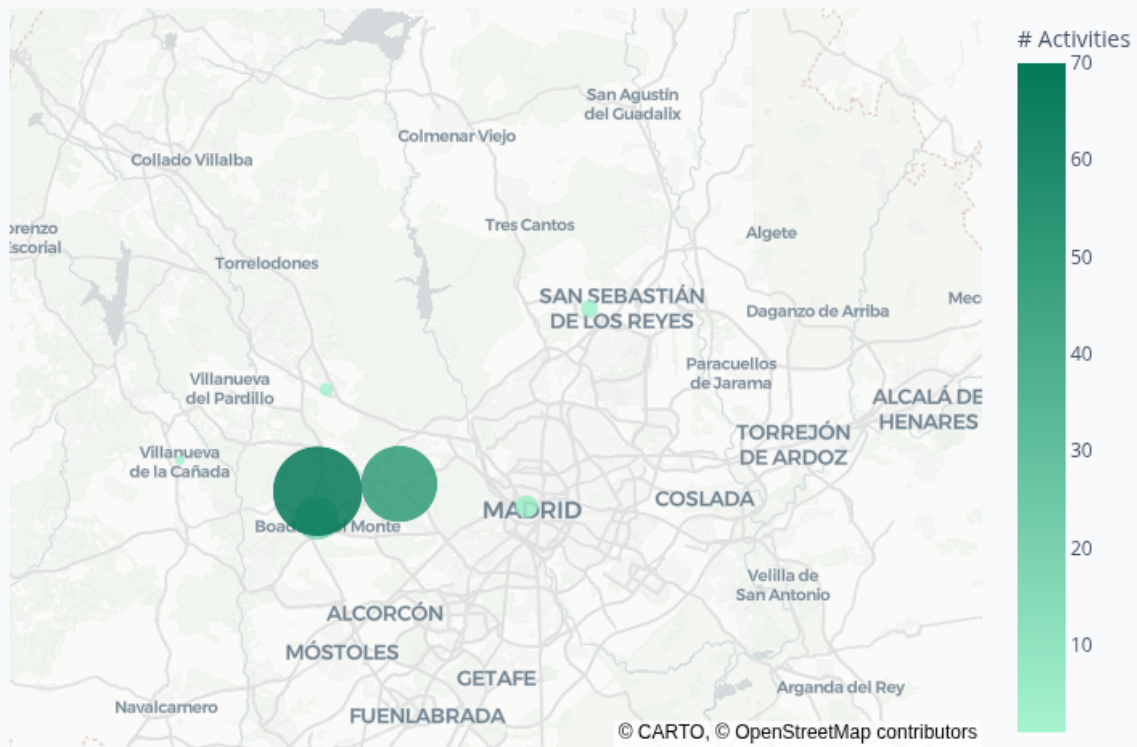


Figure 3: Q2b Chart - Activities by Municipality (Madrid)

Insights: Fig. 3 shows how most activities are centered around two different municipalities whilst some activities are recorded in other municipalities around the north west parts of Madrid, with one outlier being in Rascafría.

4.3 Q3 - Trends

This section dives into answering Q3^{2.3.3} with creative usage of charts in order to point out exercise trends throughout time periods, years and weeks. Earlier data was cleaned to create the following columns:

- Week and Year: Extracted from date time.
- Intensity: Extracted from Avg BPM.
- WHO Minutes: Extracted from time and using intensity to calculate it.

4.3.1 Q3a: Overview Heatmap

This particular chart divides itself into three different subplots. This reveals whether consistency and volume translate to outcomes.

Period Timeline This first chart grabs different activity periods which are periods during which activities are recorded with less than 60 days between each, allowing to see how consistent the exercise recording process has been. A horizontal bar chart was utilized in a creative way, displacing horizontal bars to emulate a **timeline** chart.

Frequency Heatmap A **heatmap** was chosen for this chart with a green color gradient showcasing weeks during which activities were recorded, aggregating data by week and year.

Benefits Heatmap This last chart was also a **heatmap**, but in this case, the WHO Minutes were filtered depending on whether or not a specific health benefit was obtained that week or not, allowing us to see whether activity frequency translated to health benefits.

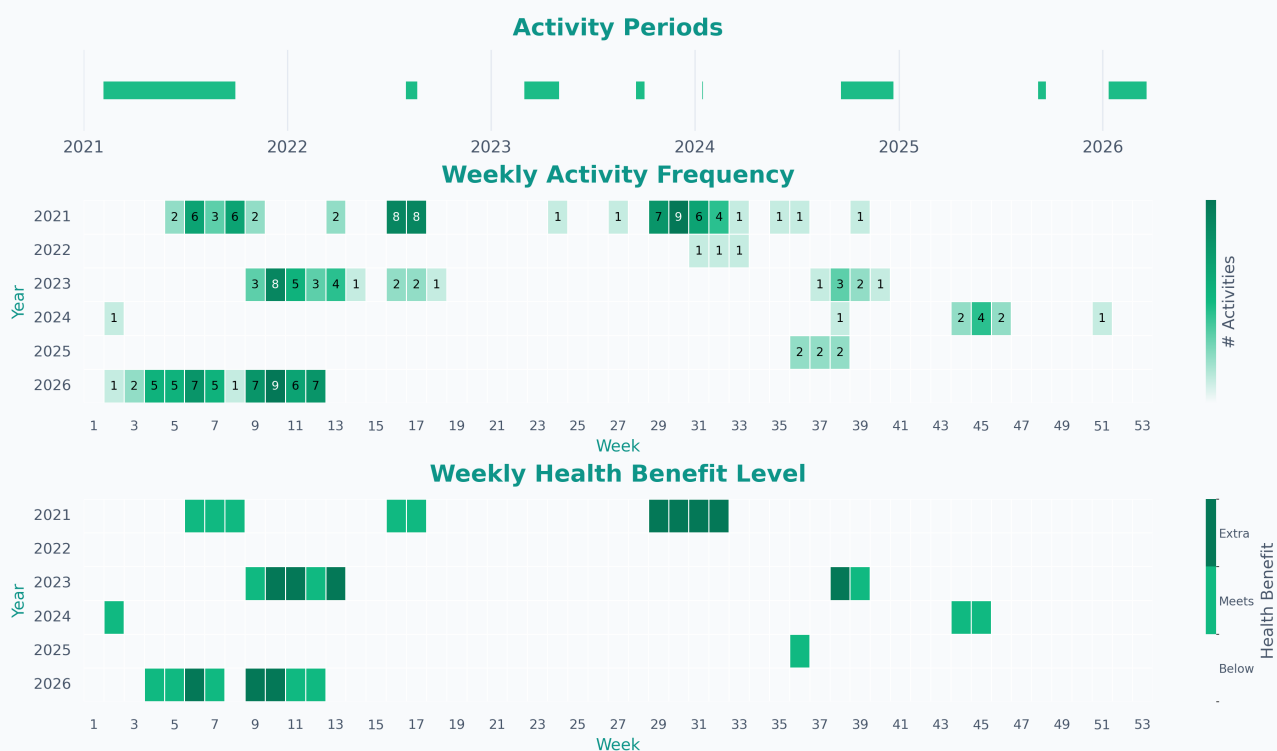


Figure 4: Q3a Chart - Overview Heatmap

Insights: Fig. 4 shows how most activities were recorded in *bursts* rather than consistently throughout the years. Specific periods are highlighted, with a current one in 2026. Most active weeks translate to health benefits as shown when comparing middle and bottom visualizations.

4.3.2 Q3b: Year-Specific Trends

Having seen the broad picture of the trend throughout the years, a zoomed in chart follows, providing a year-specific view of how activities trended. Due to a year having 53 weeks, a **lollipop** chart was chosen as opposed to a bar chart, due to the chart being too visually cluttered if the latter option was chosen.

An interactive widget was coded in to facilitate the same chart being able to be used for different years depending on what year is selected in the drop down menu. Multiple thresholds were introduced, to keep the idea of different health benefits being met if different minute amounts of activity were achieved per week, these are easily seen due to color changing as thresholds are met.

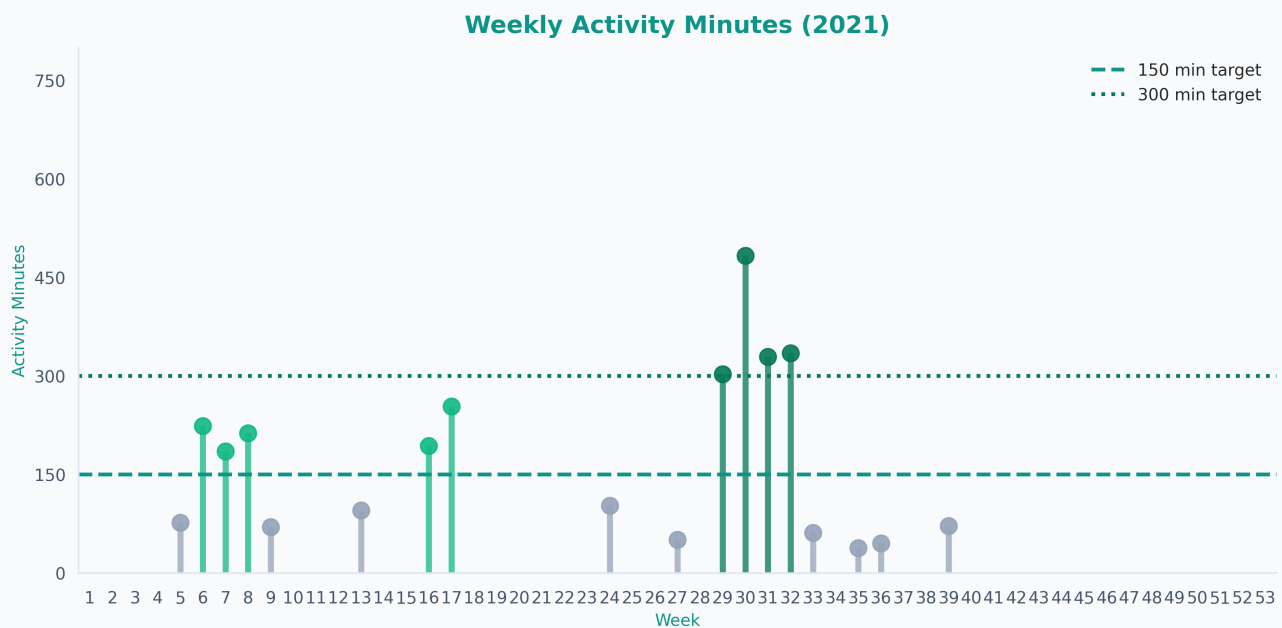


Figure 5: Q3b Chart - 2021 Activity Lollipop Chart

Insights: Fig. 5 Shows how in the year 2021 most weeks are empty due to no activities being recorded, but weeks with recorded activities tend to always have some kind of health benefit.

4.3.3 Q3c: 2026 Weekly Activity Trend

Zooming into the most recent period, a **scatter plot** with a **regression line** was chosen to show how current activity levels are evolving over time, and whether the recent trajectory suggests improvement.

As in previous charts, threshold lines are included to indicate whether weekly activity reaches different health-benefit levels. Point colors change depending on whether those thresholds are met. A confidence interval is also shown around the regression line, making it easier to judge how closely the observed weekly values follow the overall trend.

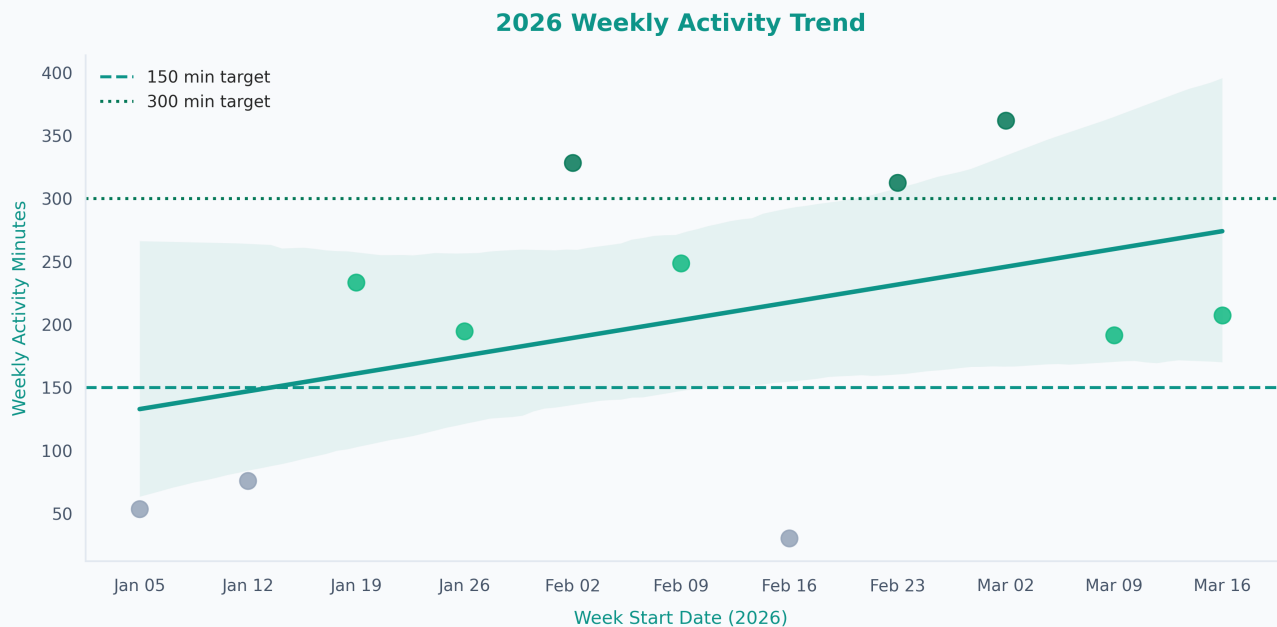


Figure 6: Q3c Chart - 2026 Weekly Activity Trend

Insights: It is still too early to draw strong conclusions, but Fig. 6 suggests a slight upward trend in weekly activity. More recent points are also more likely to reach at least one health benefit threshold than the earlier ones.

4.3.4 Visualization Summary

Question	Charts	Chart Type	Library
Q1 ^{2.3.1}	Fig.1	Horizontal bar	Matplotlib
Q2 ^{2.3.2}	Fig.2	Choropleth	Plotly
	Fig.3	Scattermap	Plotly
Q3 ^{2.3.3}	Fig.4	Timeline + Heatmap	Matplotlib + Seaborn
	Fig.5	Lollipop (interactive)	Matplotlib + ipywidgets
	Fig.6	Scatter + Regression	Matplotlib + Seaborn

Table 1: Visualization summary table

Table 1 summarizes:

- 8 chart types: Horizontal bar, choropleth, scattermap, heatmap, timeline, lollipop, scatter and regression.
- Interactive and static visualizations.
- 4 libraries used: Matplotlib, Plotly, Seaborn and ipywidgets.

When it comes to datasets, even though there was one dataset in csv format, the resulting dataframe was transformed into specific datasets for each different chart, showing how one dataset can be used in many different creative ways.

5 Conclusions

Visualizations reveal the following:

- Running and walking dominate activities [Fig. 1].
- Most activities recorded in Madrid, around a specific area [Fig. 2].
- Activities recorded in bursts (periods) [Fig. 4].
- WHO target for 150 minutes met when activities were recorded, 300+ minutes one met sporadically [Fig. 5].
- Current recorded period shows upward trend [Fig. 6].

6 References

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